

Ethanol Blending in Gasoline - Argentina

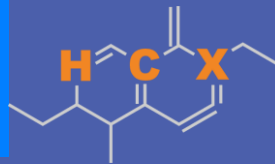
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



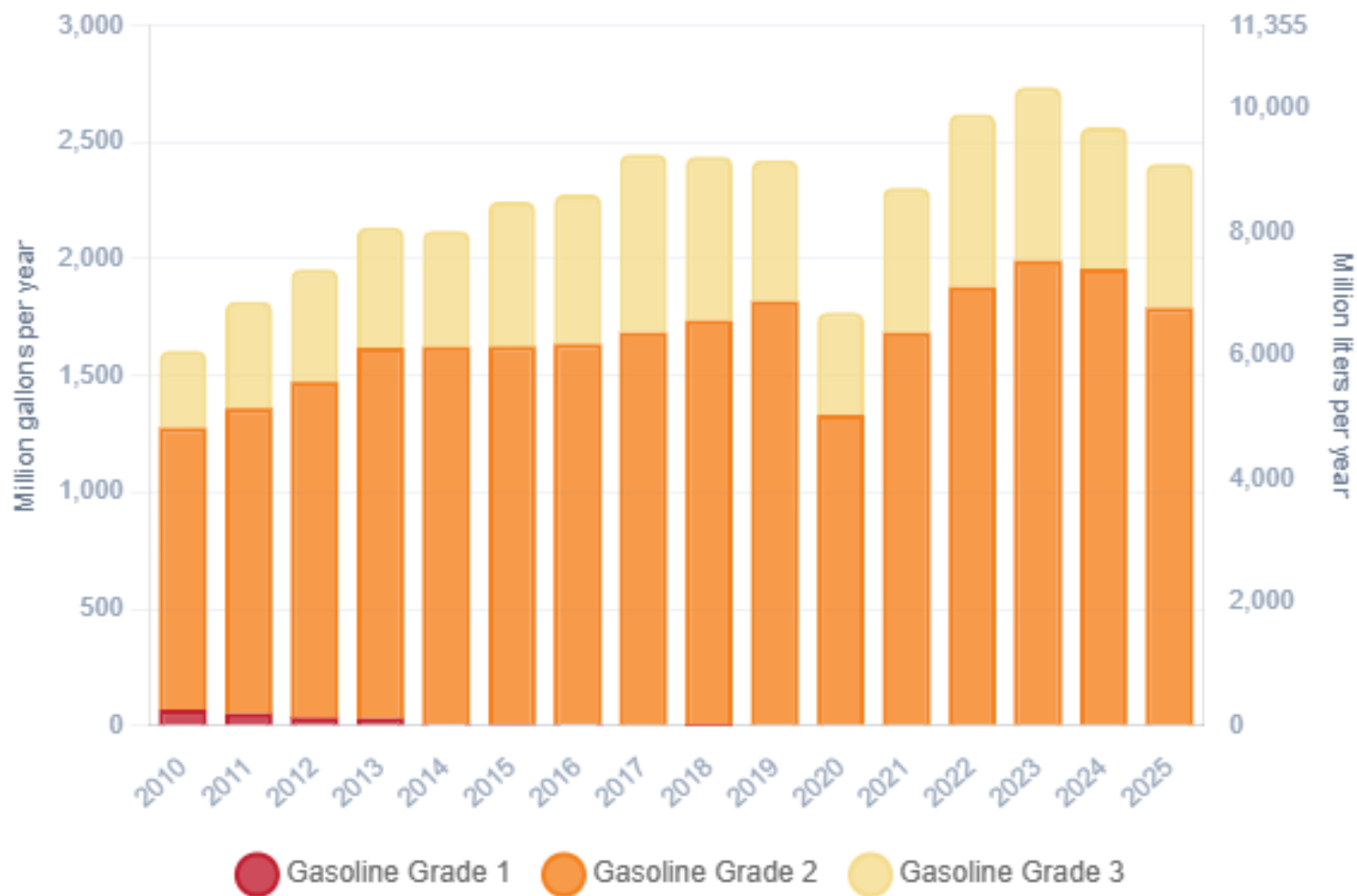
Ethanol Blending in Gasoline - Argentina



Argentina's total gasoline demand was 2,408 million gallons in 2025 (9,117 million liters), with a market share of approximately 74.2% as Nafta Súper / Grado 2 (RON 93 min – AKI 87) and 25.8% as Nafta Ultra / Grado 3 (RON 97 min – AKI 92). Octane delivered to market is usually higher than the minimum required. The demand for gasoline is supplied mostly by local production, requiring imports of less than 75 million liters per year average.

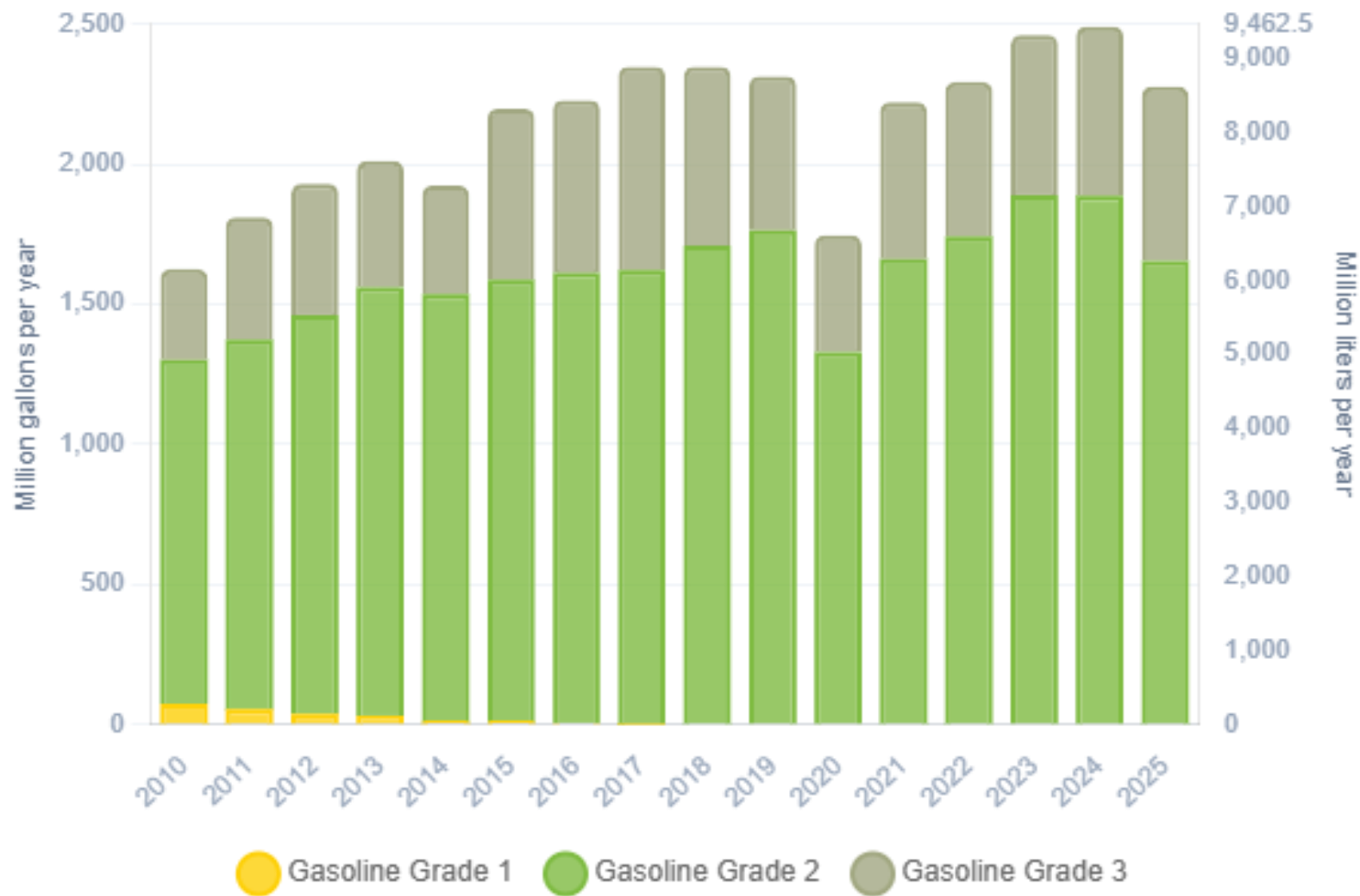
Argentina's policy is self-sufficiency and production and supply of ethanol for gasoline blend. Ethanol concentration varies according to local production. Currently, the concentration limit is 15% v/v. (Resolution 2026-79-APN-SE#MEC). Ethanol production is equal to consumption with some exports and imports from other countries.

Gasoline Demand in Argentina



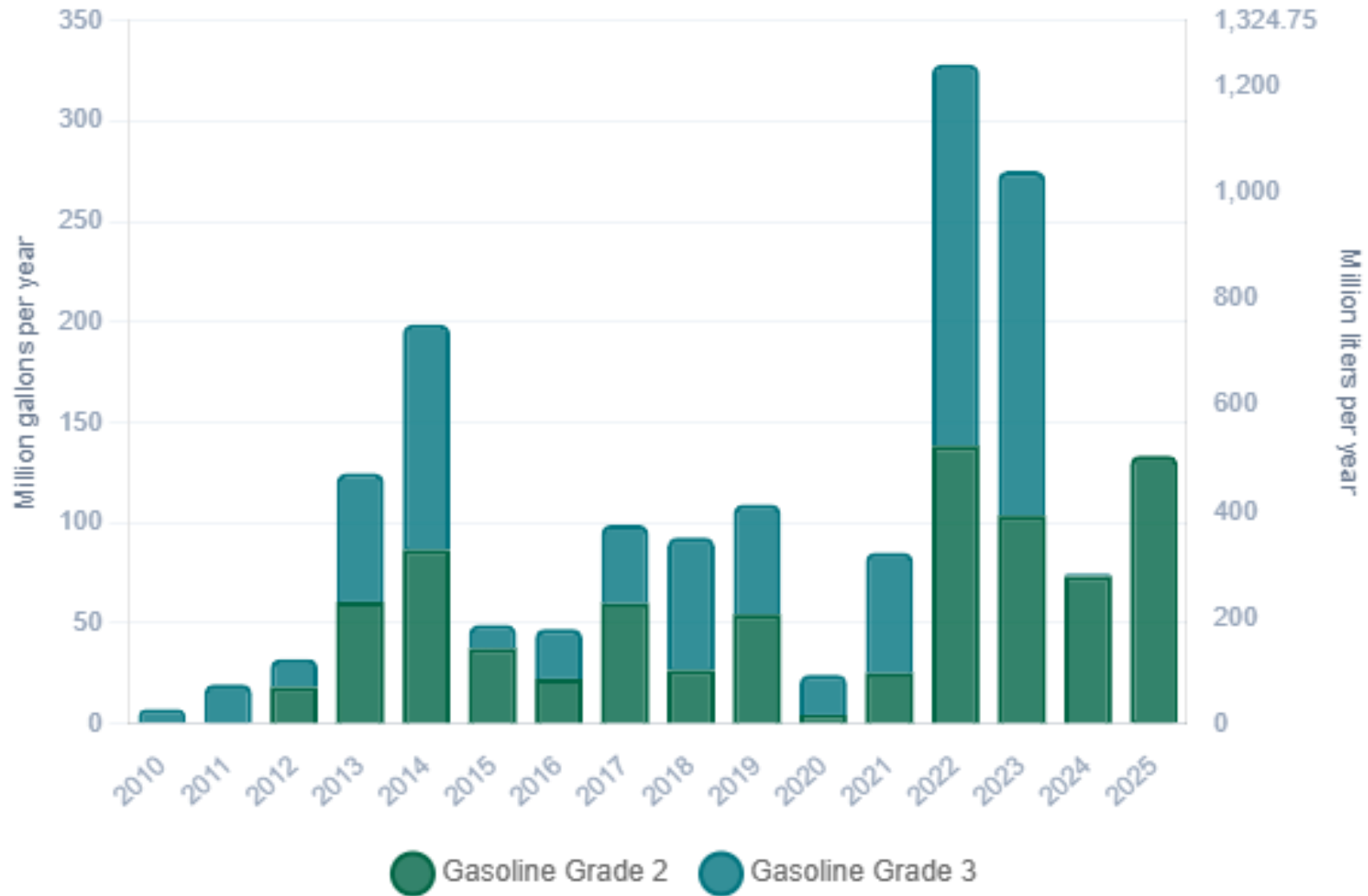
Source: Ministerio de Economía, 2025

Gasoline Production in Argentina



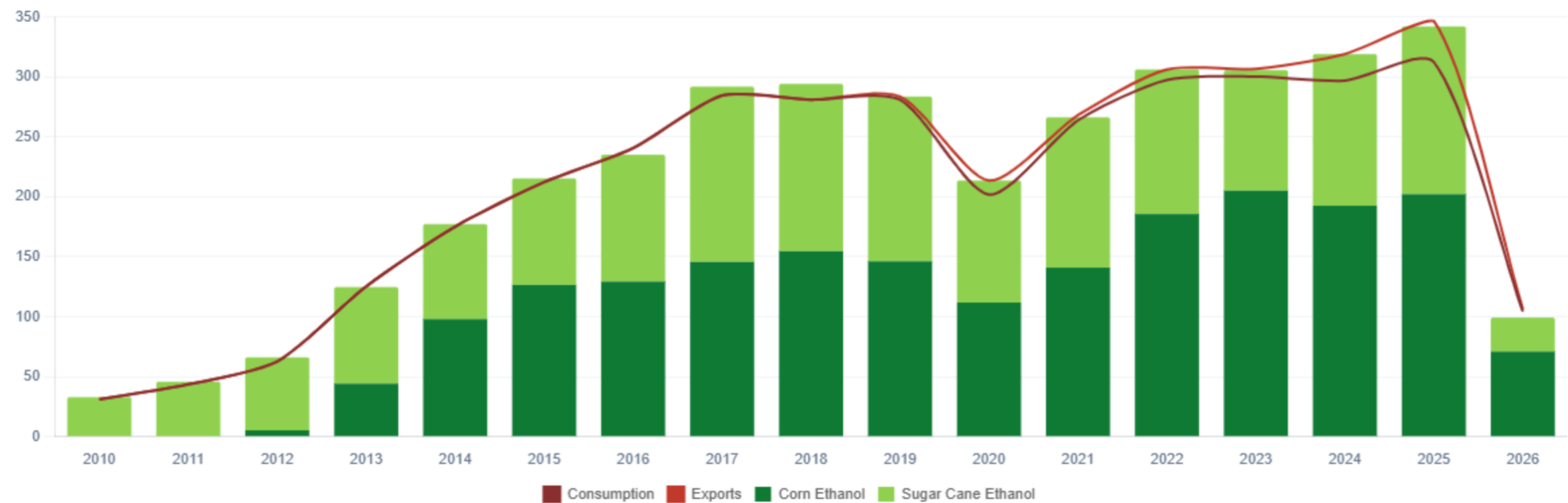
Source: Ministerio de Economía, 2025

Gasoline Imports in Argentina



Source: Ministerio de Economía, 2025

Ethanol Balance in Argentina



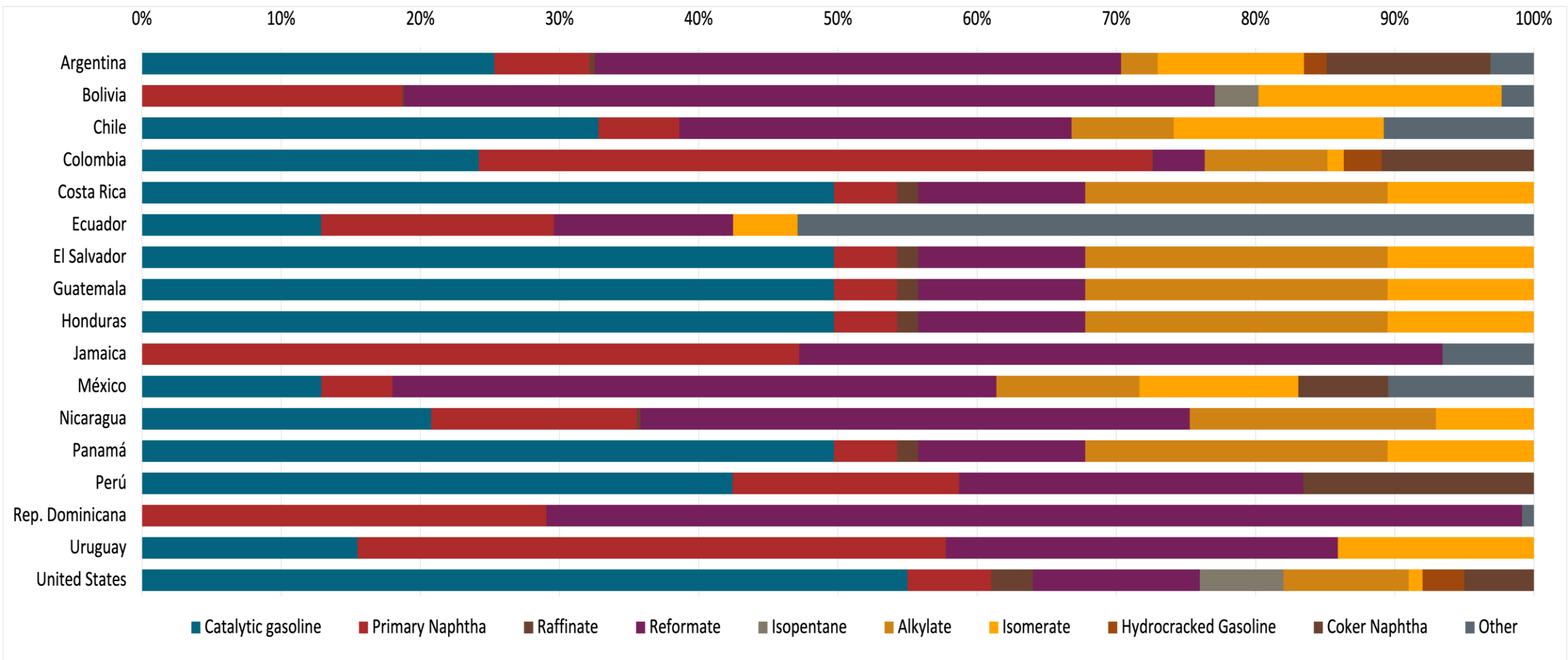
Source: Ministerio de Economía, 2025

Gasoline Quality in Argentina

Name	Resolution 576/2019 and Resolution 492/2023						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019						2017			
Applicability	North Zone		Central Zone		South Zone		All countries			
Selected Grade	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 93	> 97	> 93	> 97	> 93	> 97	> 95	> 95	> 98	> 98
MON	> 83	> 85	> 83	> 85	> 83	> 85	> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa				
RVP 37.8°C (Transition)			<> 45 - 80 kPa	<> 45 - 80 kPa						
MTBE	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	-	-	-	-
Ethers 5 or more C Atoms	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



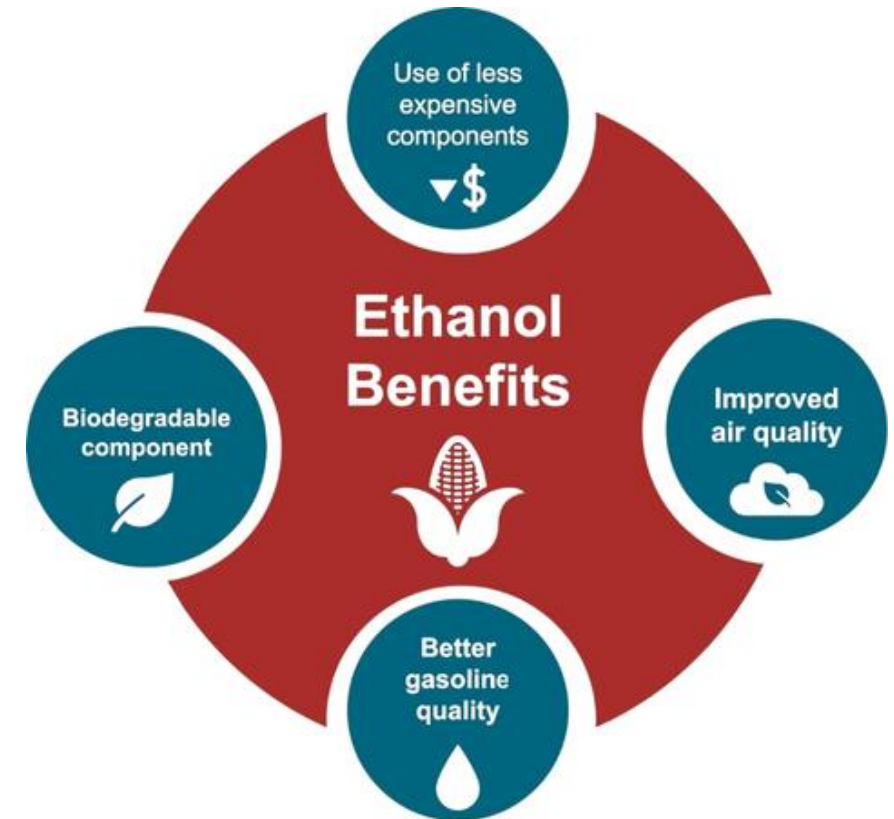
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

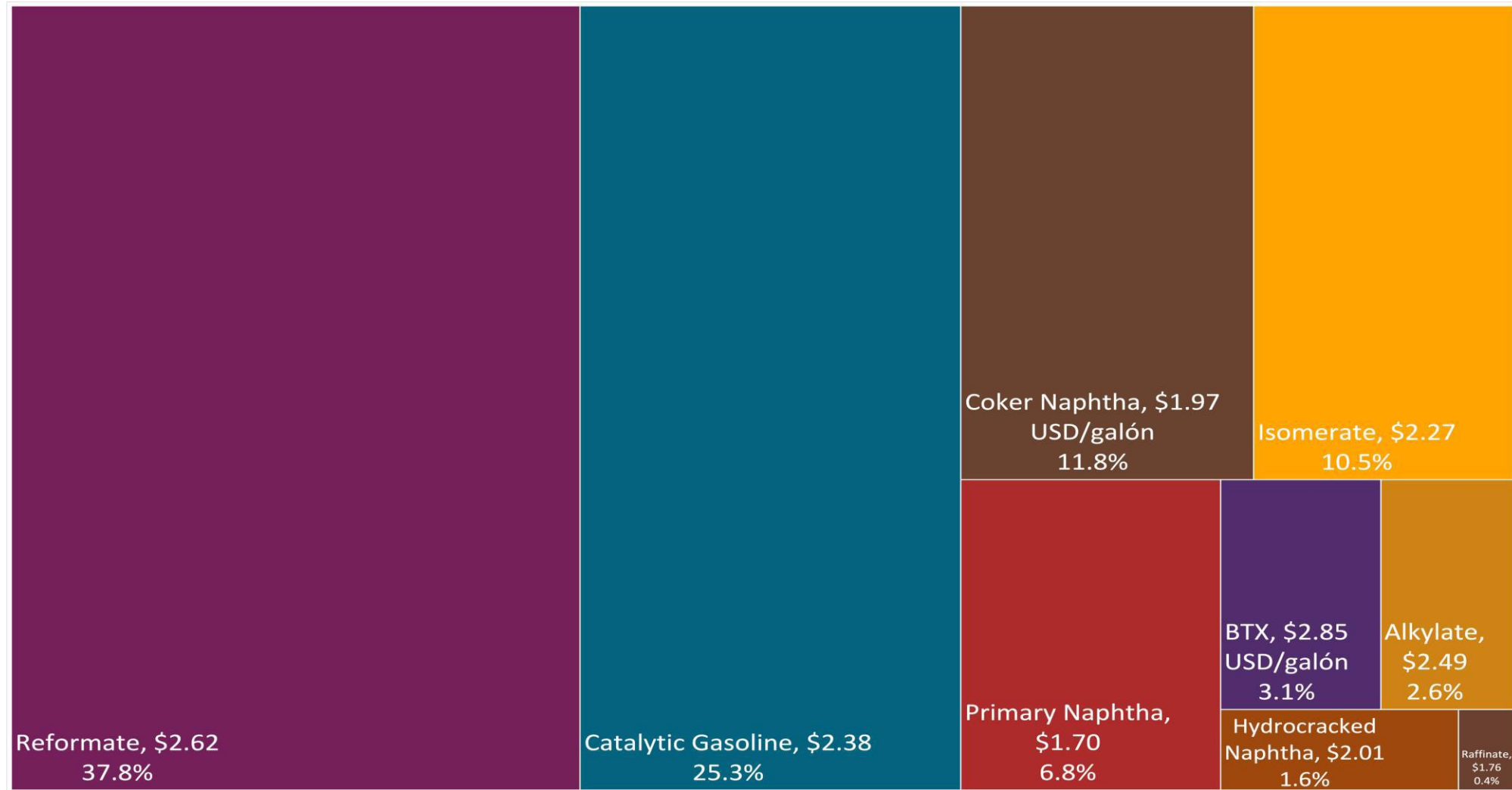
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

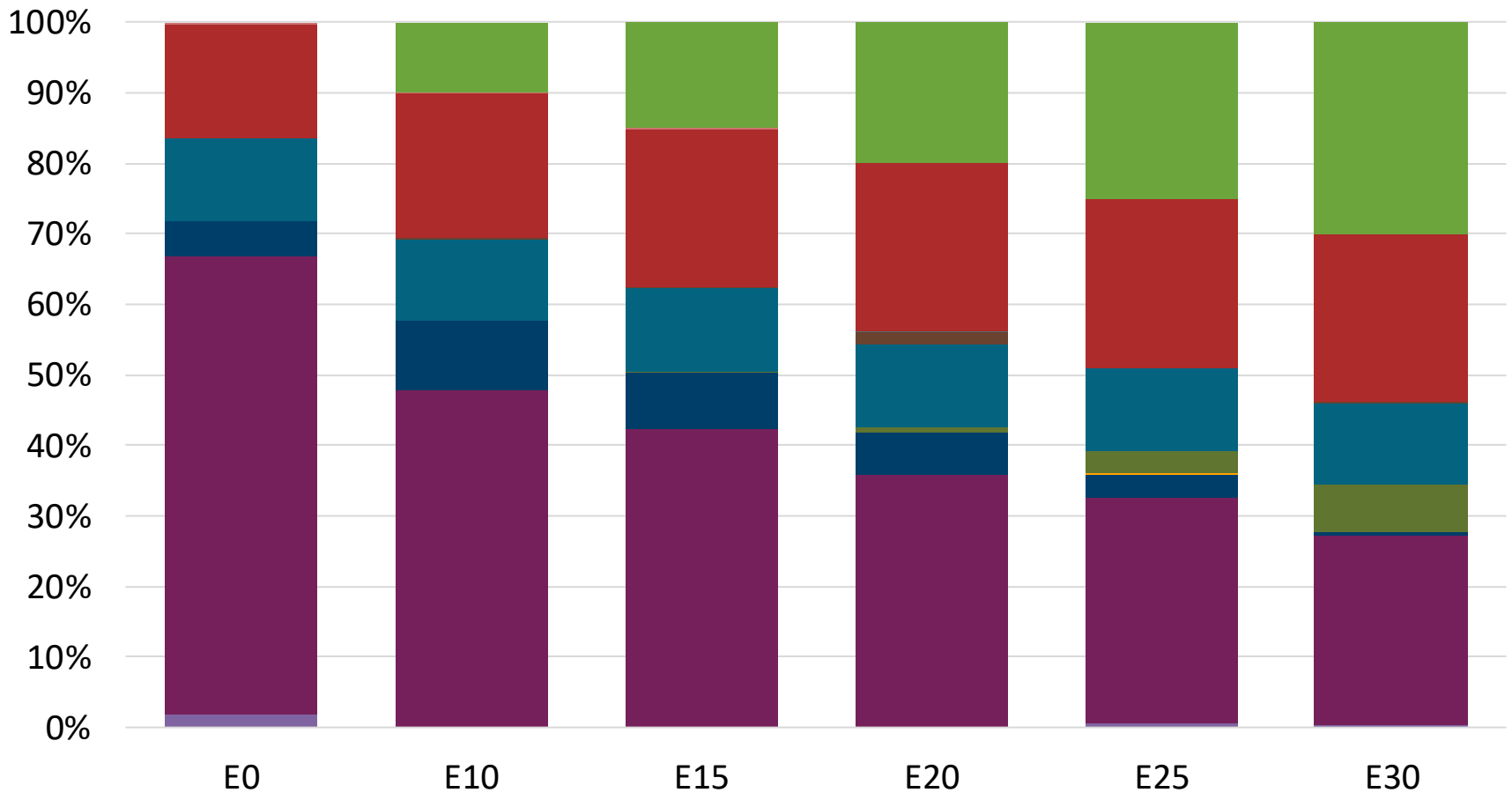
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

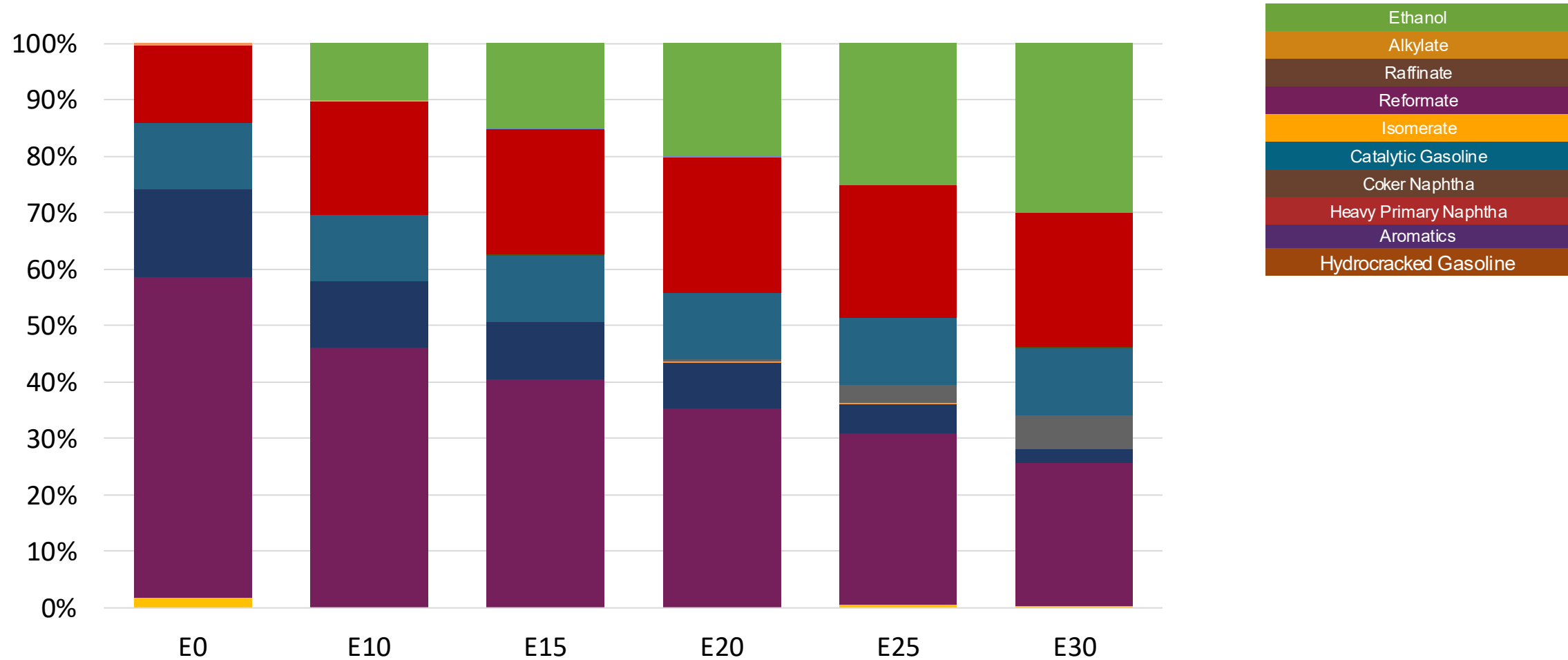
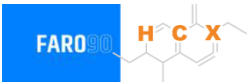


Argentina – Nafta Grado 2 – North & Central Zones– Constant Octane



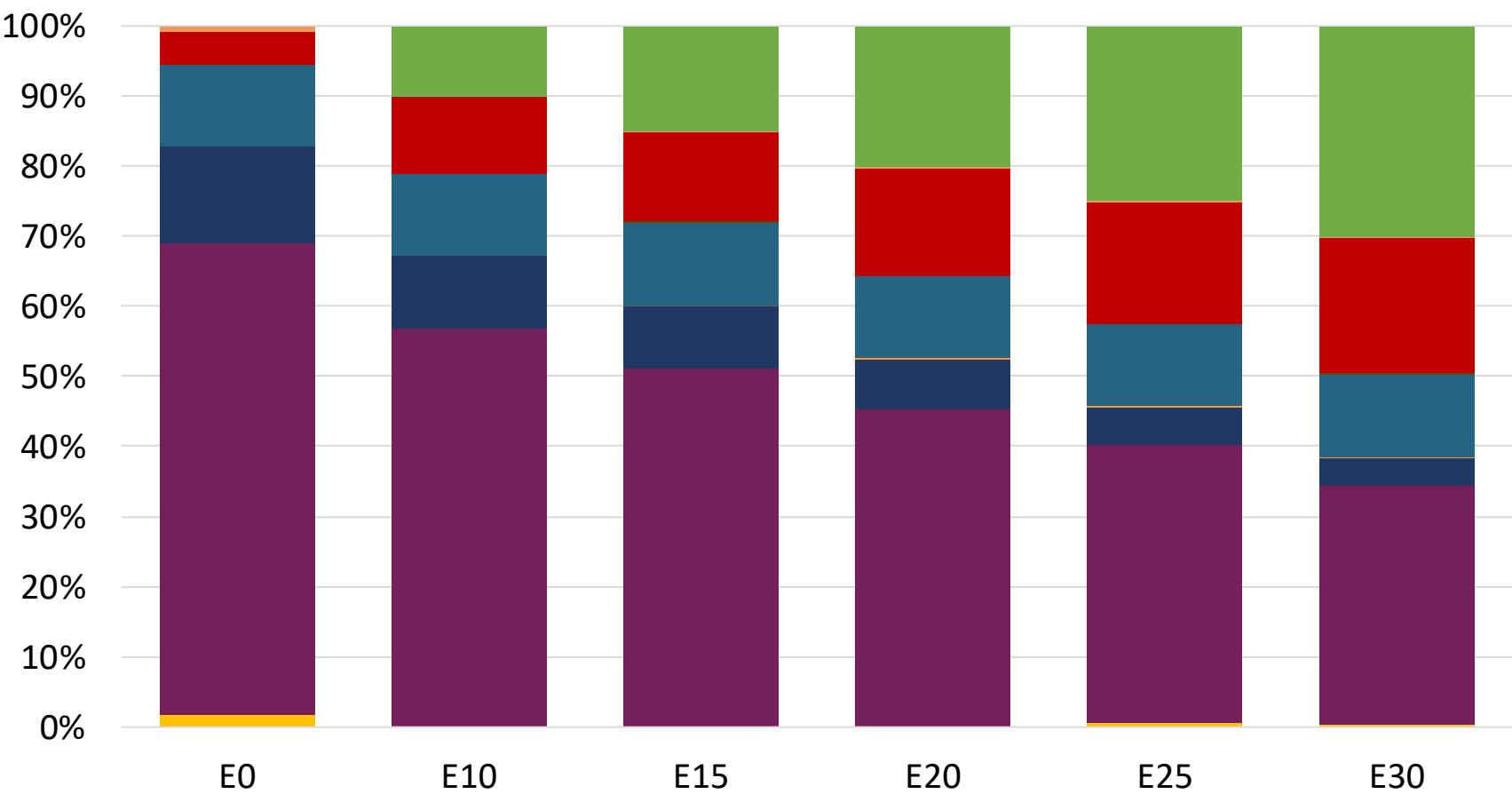
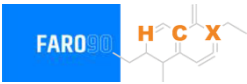
Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US\$/gal)	\$ 2.07	\$ 1.83	\$ 1.77	\$ 1.70	\$ 1.64	1.58

Argentina – Nafta Grado 2 – South Zone – Constant Octane



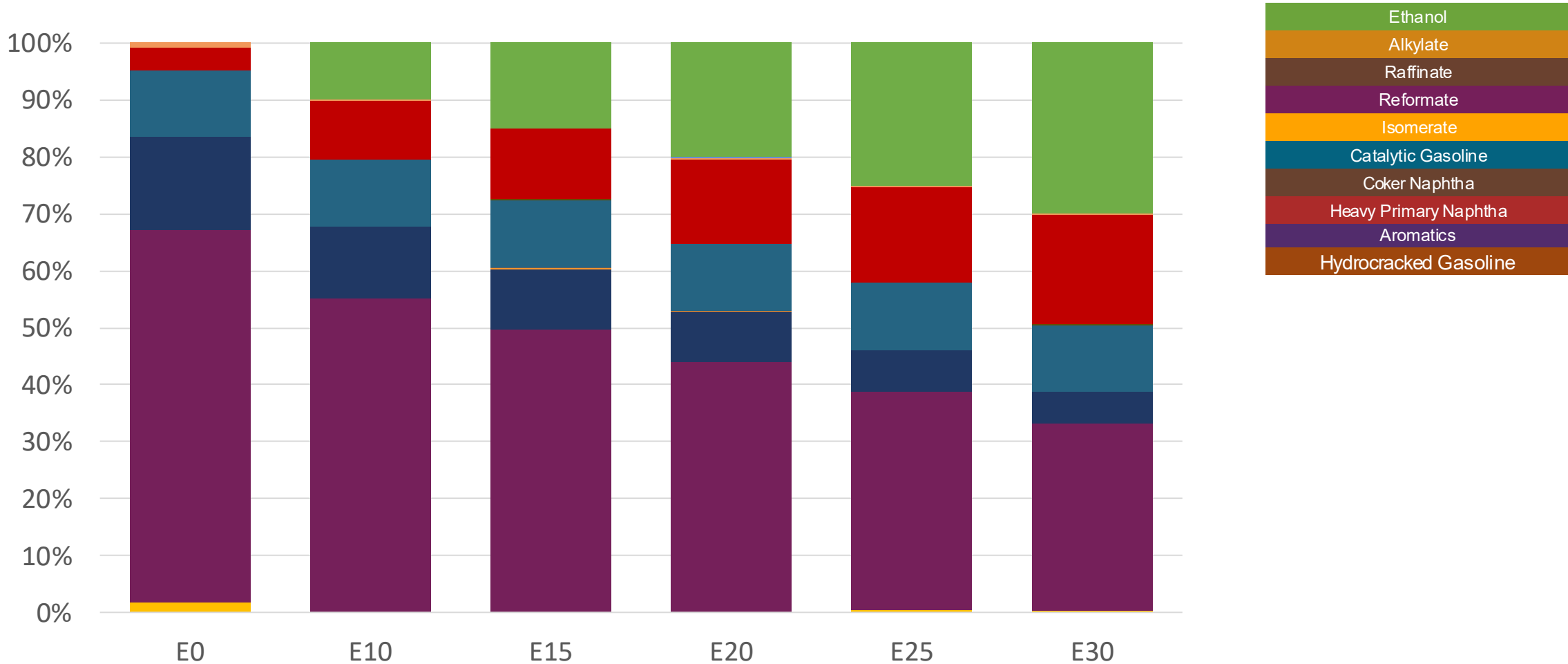
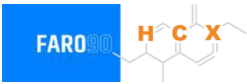
Octane (RON)	E0	E10	E15	E20	E25	E30
Price (US\$/gal)	\$ 1.92	\$ 1.80	\$ 1.74	\$ 1.68	\$ 1.62	1.56

Argentina – Nafta Grado 3 – North & Central Zones– Constant Octane



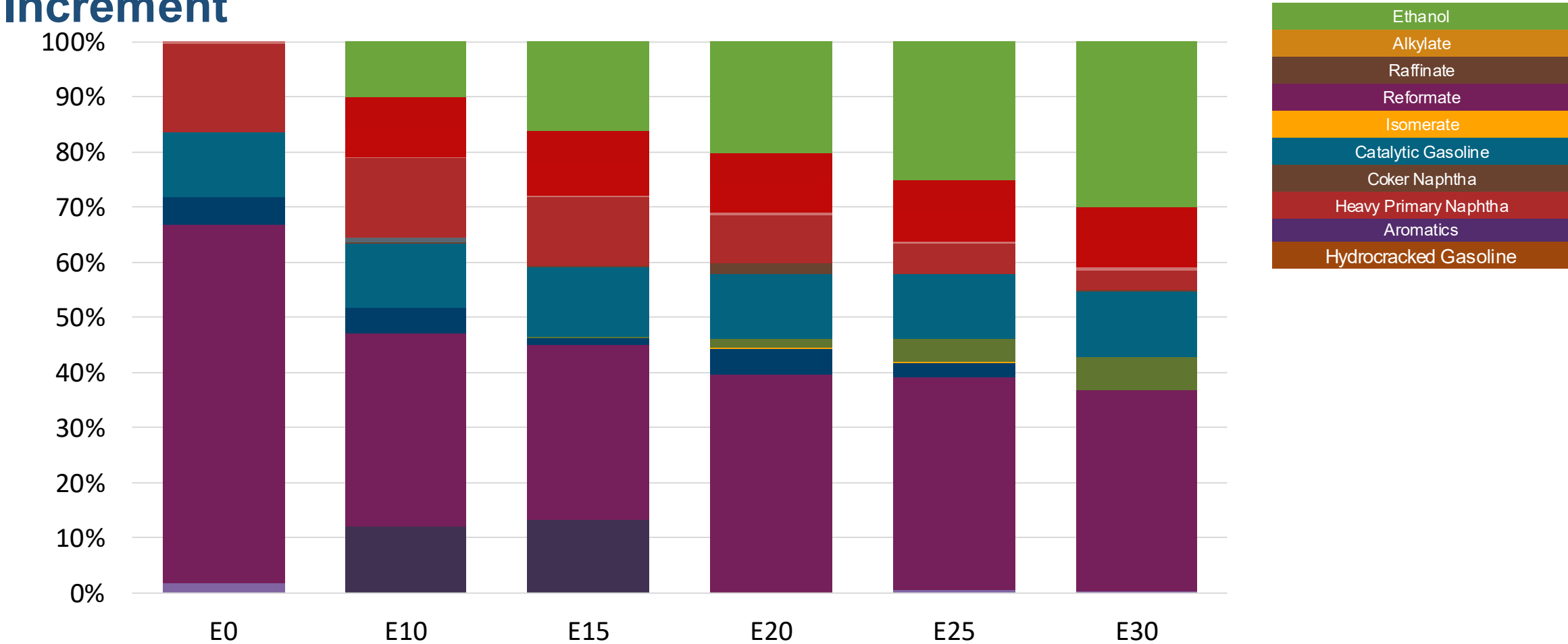
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.32	\$ 2.19	\$ 2.13	\$ 2.07	2.01	\$ 1.95

Argentina – Nafta Grado 3 – South Zone – Constant Octane



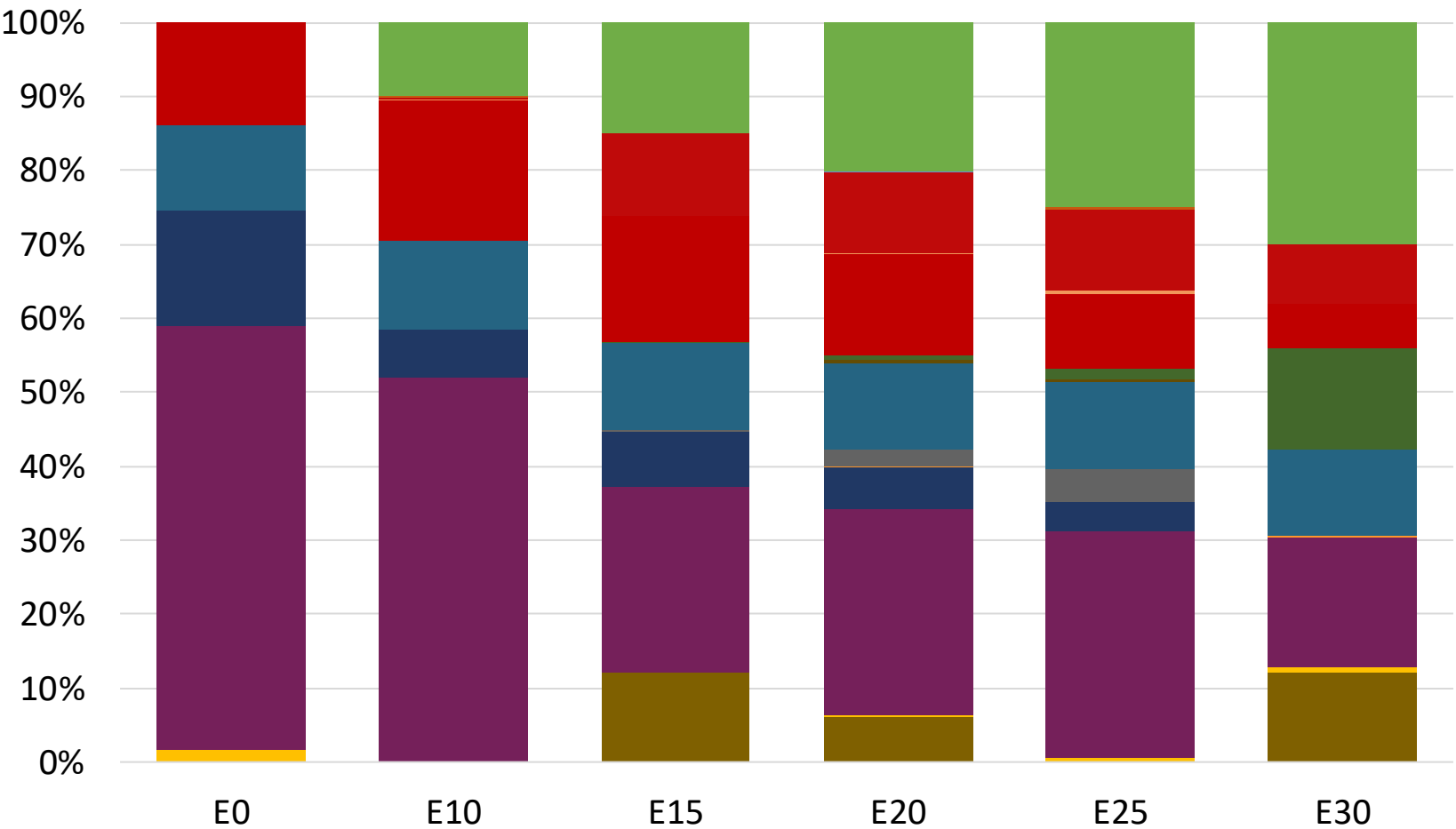
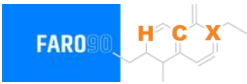
Octane (RON)	97.0		97.0		97.0	
Price (US\$/gal)	\$	2.11	\$	1.98	\$	1.73

Argentina – Nafta Grado 2 – North & Central Zones– Octane Increment



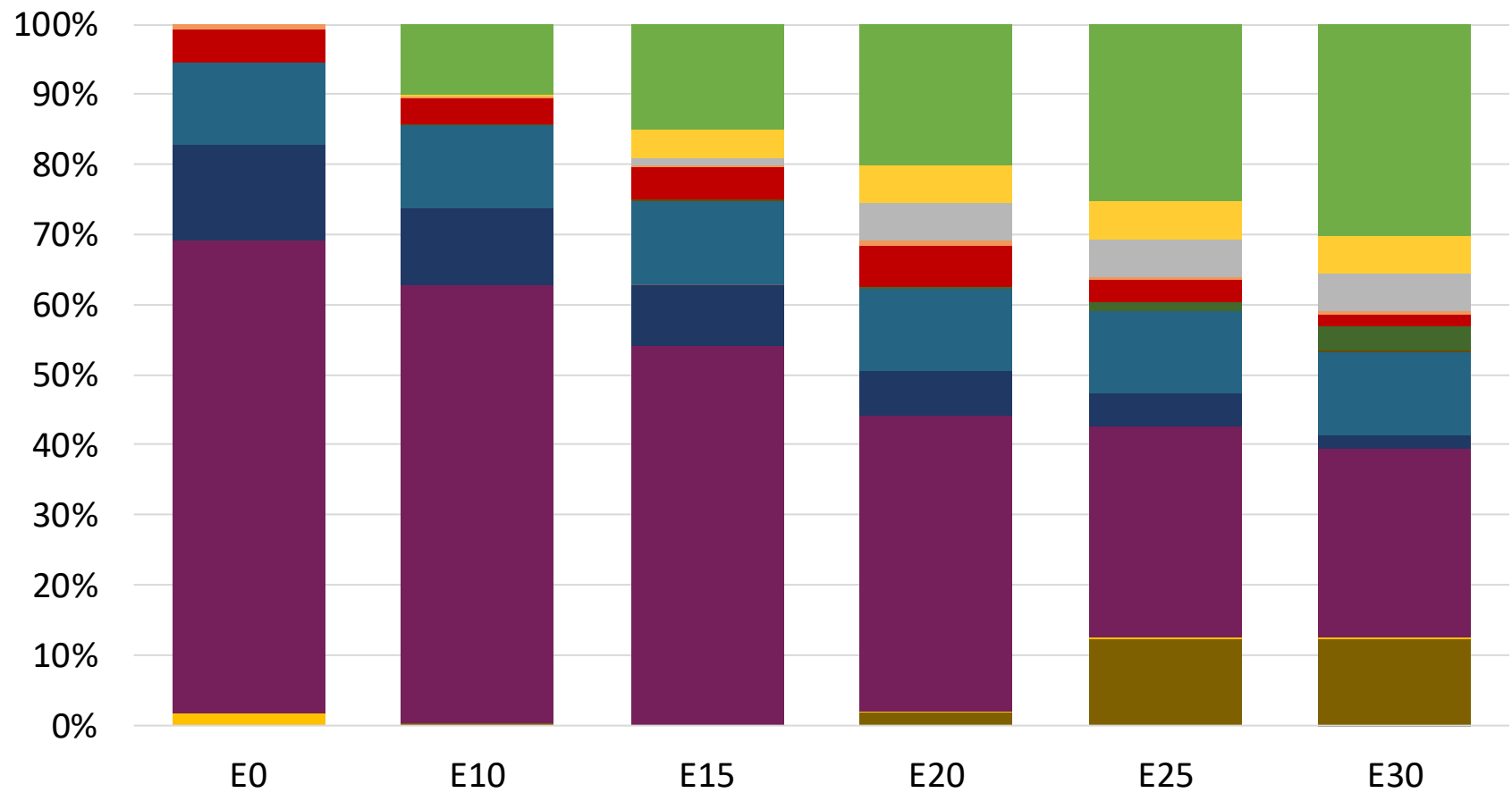
Octane (RON)	93.0	95.4	96.0	99.9	101.3	102.6
Price (US\$/gal)	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07

Argentina – Nafta Grado 2 – South Zone – Octane Increment



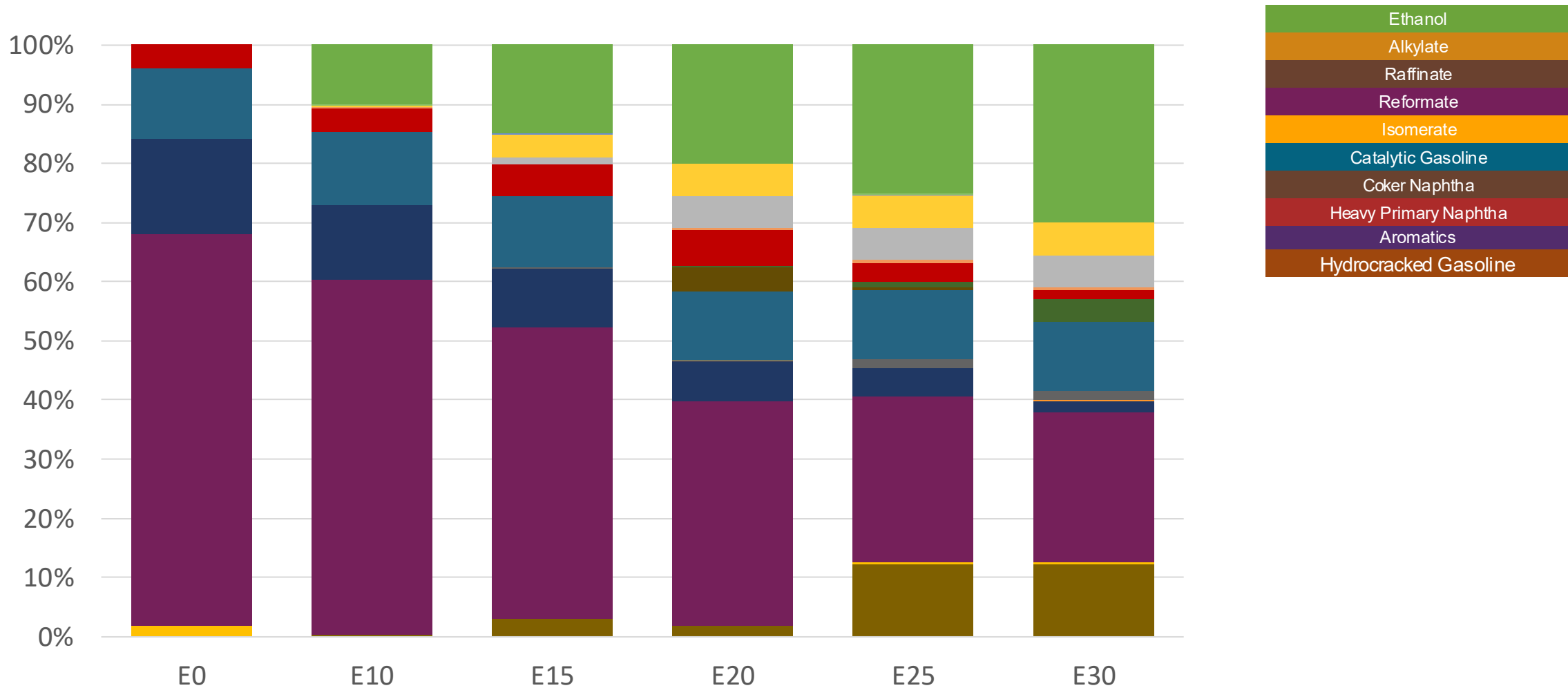
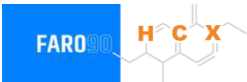
Octane (RON)	93.0	93.9	94.9	96.8	98.6	98.1
Price (US\$/gal)	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92

Argentina – Nafta Grado 3 – North & Central Zones– Octane Increment



Octane (RON)	97.0	99.8	100.8	101.6	102.2	103.2
Price (US\$/gal)	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35

Argentina – Nafta Grado 3 – South Zone – Octane Increment



Octane (RON)	97.0	99.5	100.2	100.9	101.6	102.6
Price (US\$/gal)	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

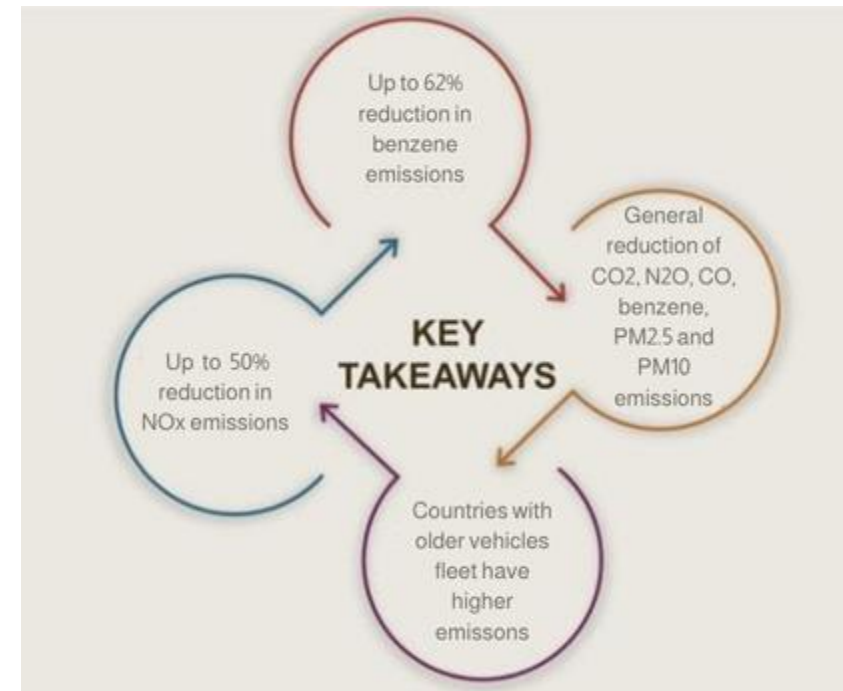
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

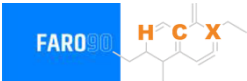
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Argentina

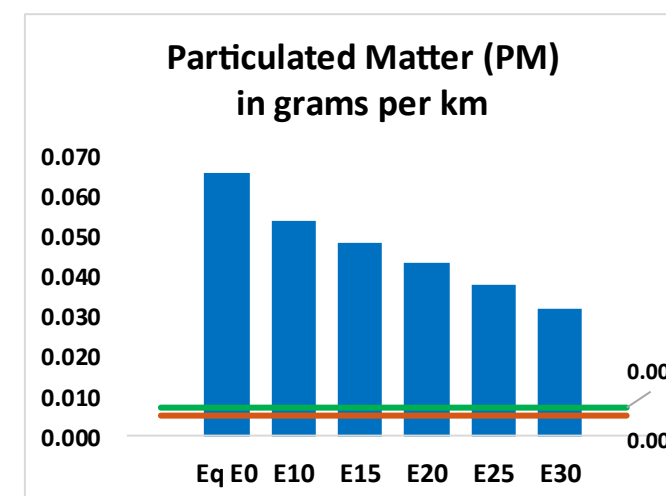
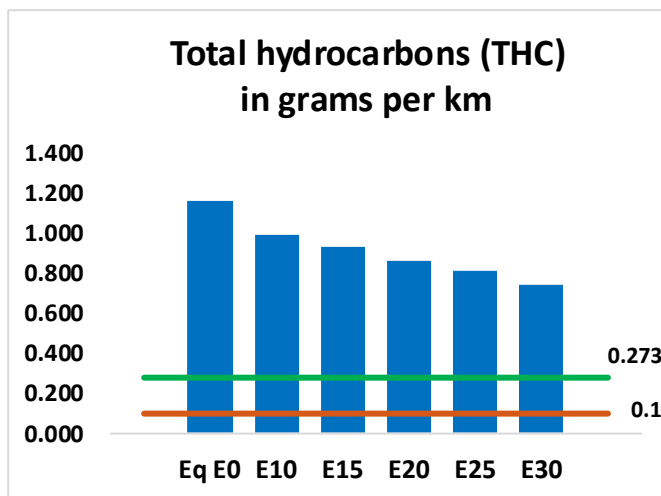
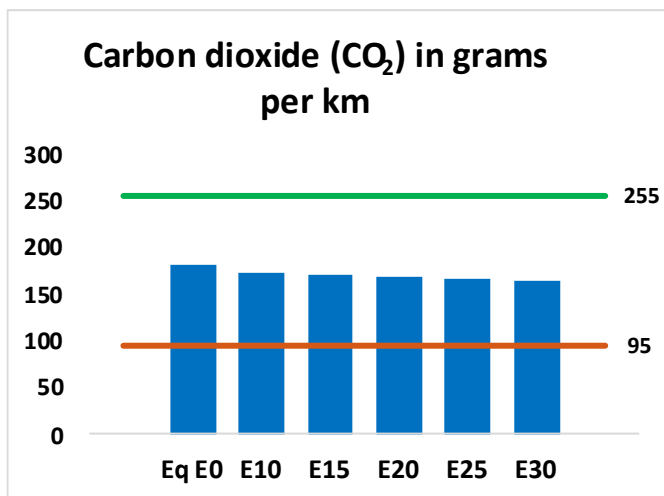
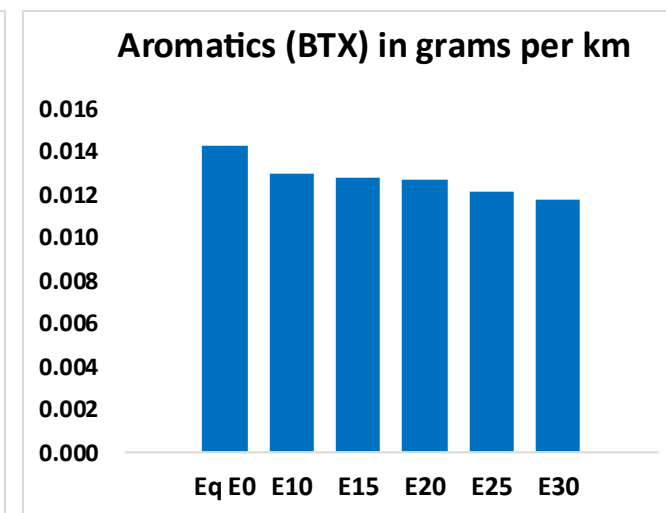
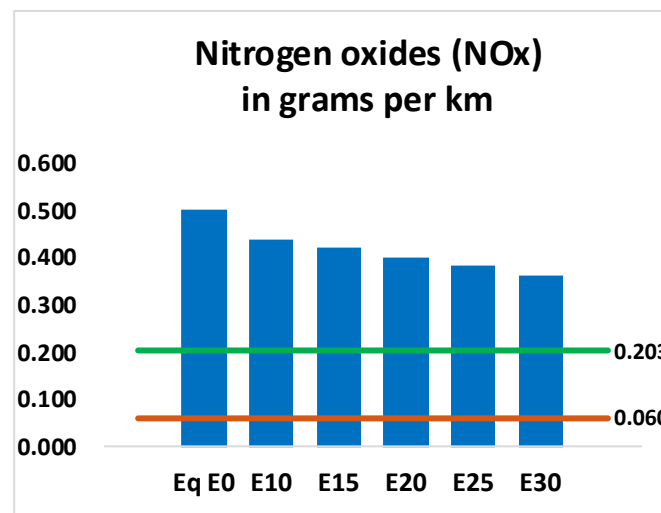
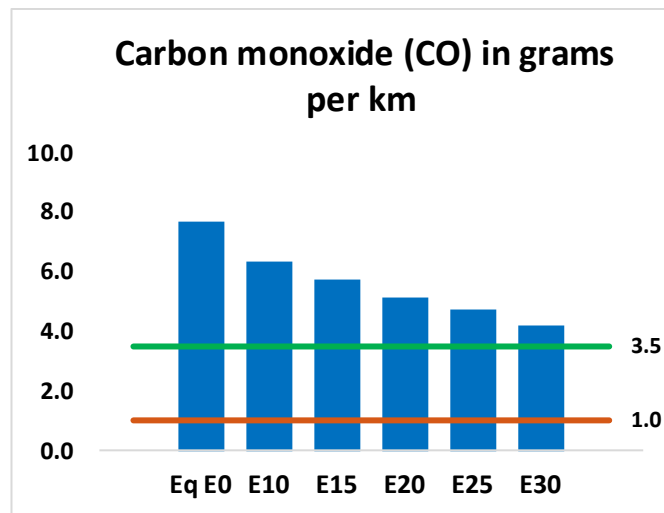


Vehicular Fleet: **17,673,409**
Average age: **12,5 years**
Motorcycles: **41.6%**

Source: Ministerio de Transporte, CAFAM, AFAC, 2025

Argentina – Vehicular Emissions

— United States
— Euro 6 Standard



Argentina – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,527,943	2,303,930	2,079,917	1,890,558	1,697,335	1,547,954	1,389,122	-18%	-33%
CO2	59,981,136	58,438,764	56,982,080	55,836,440	55,269,533	55,042,406	54,027,816	-5%	-8%
VOC	381,844	354,527	327,209	305,221	283,277	266,458	243,820	-14%	-26%
NOx	165,413	152,732	143,910	139,352	132,081	126,349	119,620	-13%	-20%
SOx	680	628	577	533	492	447	399	-15%	-28%
NH3	26,210	25,951	25,691	25,813	26,104	26,308	26,477	-2%	0%
N2O	1,105	1,099	1,094	1,100	1,122	1,135	1,148	-1%	2%
THC	83,875	83,875	83,875	85,133	86,979	88,572	90,082	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,780	2,613	2,446	2,319	2,195	2,101	1,971	-12%	-21%
Acetaldehyde	7,881	9,045	13,272	20,211	27,536	31,465	37,179	68%	249%
Formaldehyde	31,273	30,404	35,442	41,617	43,600	48,233	52,507	13%	39%
Benzene	4,695	4,511	4,275	4,207	4,172	3,990	3,861	-9%	-11%
PM2.5	3,991	3,628	3,265	2,946	2,633	2,299	1,948	-18%	-34%
PM10	17,625	16,023	14,421	13,011	11,631	10,152	8,604	-18%	-34%

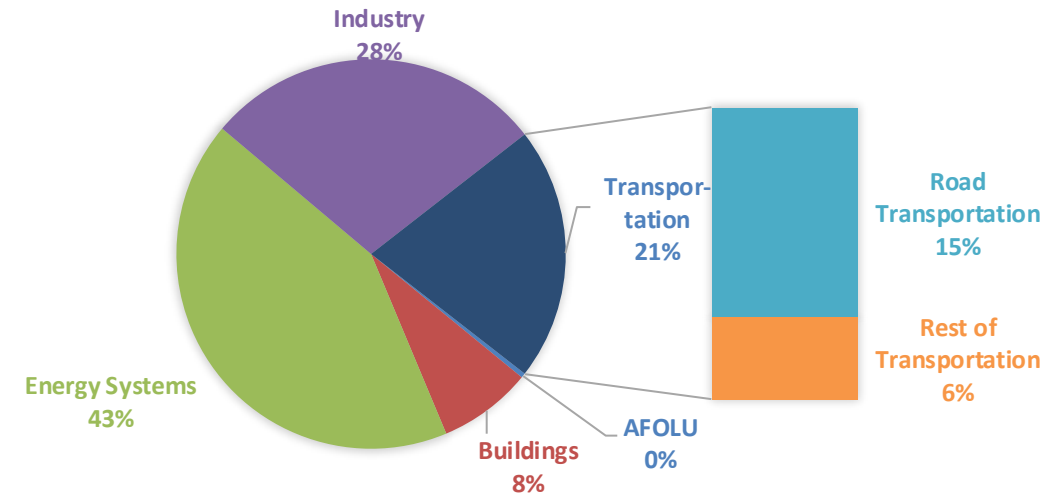
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

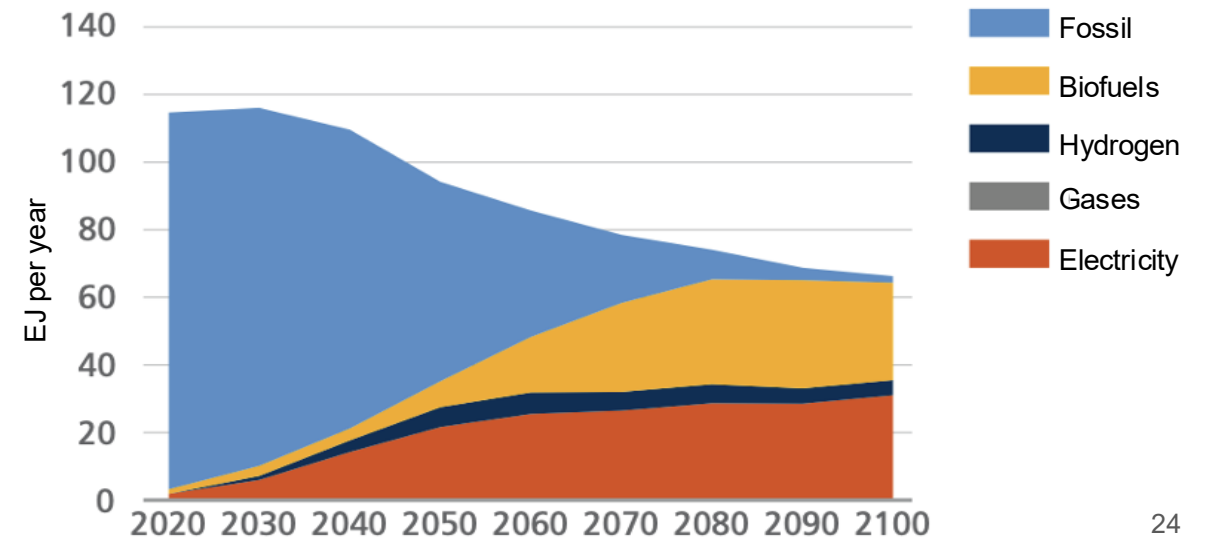
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

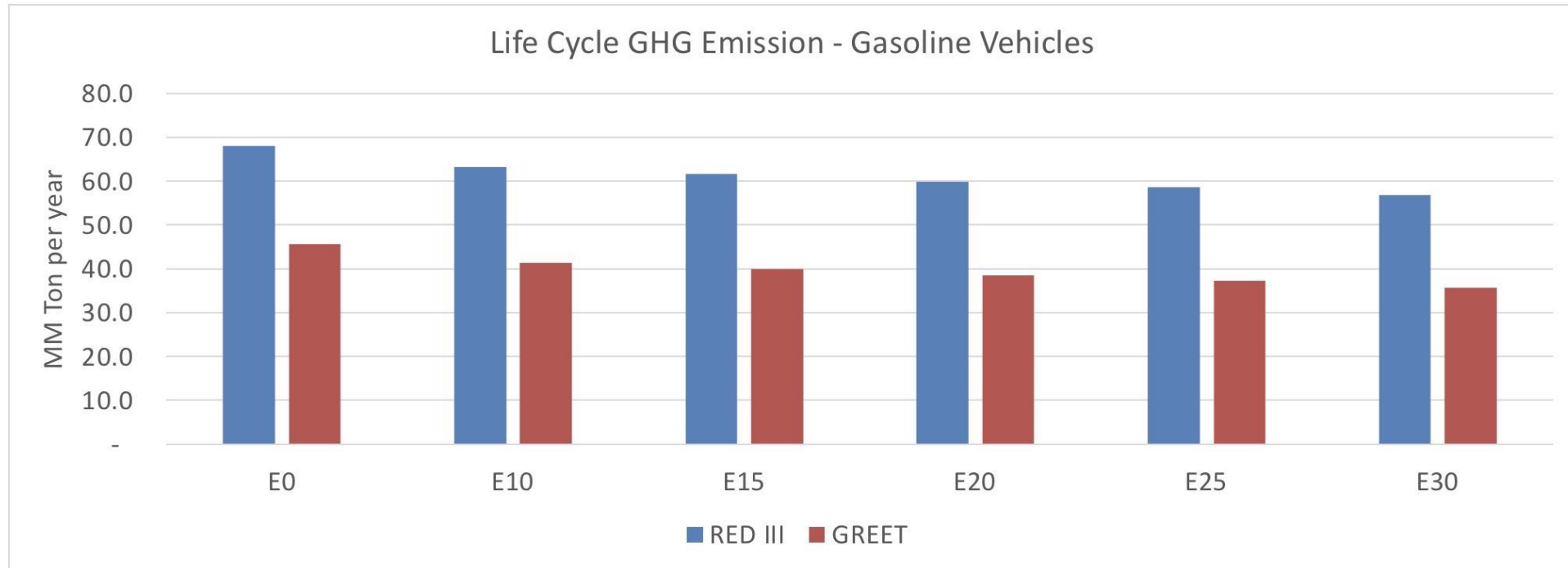
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Argentina – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/año)	359.7
Target @ 2035 (MMT/year)	197.5
Delta	162.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.4%	15.6%	18.2%	21.7%
RED II/III	0.0%	7.1%	9.5%	11.9%	14.0%	16.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	2.7%	3.5%	4.4%	5.2%	6.1%
RED II/III	0.0%	3.0%	4.0%	5.0%	5.9%	7.0%